EE4091: Photonics - Final Exam Duration: 60 minutes Solve all questions

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(1) A 0.5-μm plane light wave of intensity I_o is incident on a barrier containing two small slits of size 0.1 mm. The openings are 0.4 mm apart. Describe the interference pattern created on a screen 5 meters away? Give all details including a 1-D sketch of the pattern showing the location of the first 5 dark spots.

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(2) Explain two reasons why Si is not used for green LEDs. For each reason give a solution. Suggest an alternative material and explain why.

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(3) Give two advantages and two disadvantages of using heterojunction structures in the design of LEDs? Explain each item. Is this design more effective or less effective in high energy LEDs? Explain.

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(4) Consider two LEDs with the same responsivity; a blue LED emitting at 480 nm and a yellow LED emitting at 590 nm. If the two LEDs are to give the same brightness in a shop display, what output optical power should each produce? Which one should carry more current? Explain your answers in detail and use equations?

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(5) Explain the effect of the value of the load resistance on the conversion efficiency of a solar cell. Describe a method to determine the load resistance for optimal conversion.